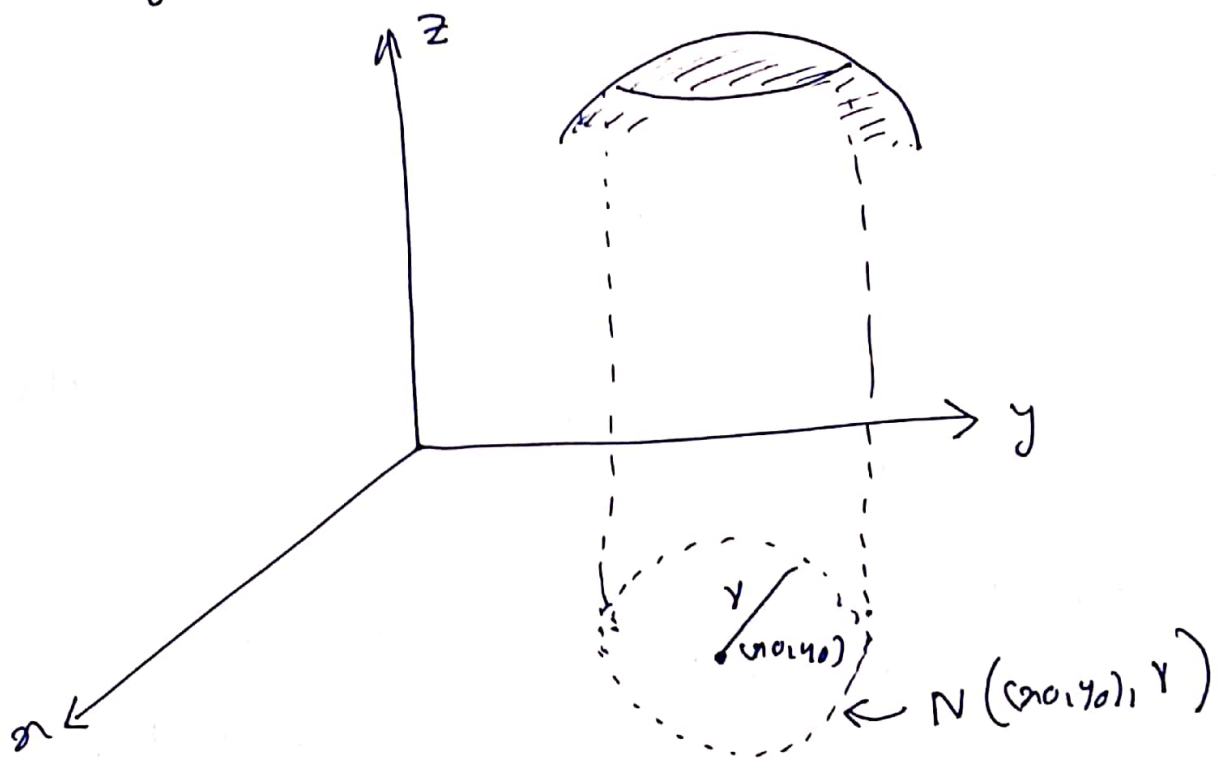


- Extreme Values of functions of two variables. -

Let f be a real valued function of two variables defined in a neighbourhood of a point (x_0, y_0) .

This function is said to have a local maximum at (x_0, y_0) if $f(x, y) \leq f(x_0, y_0)$ for all points (x, y) sufficiently closed to (x_0, y_0) .

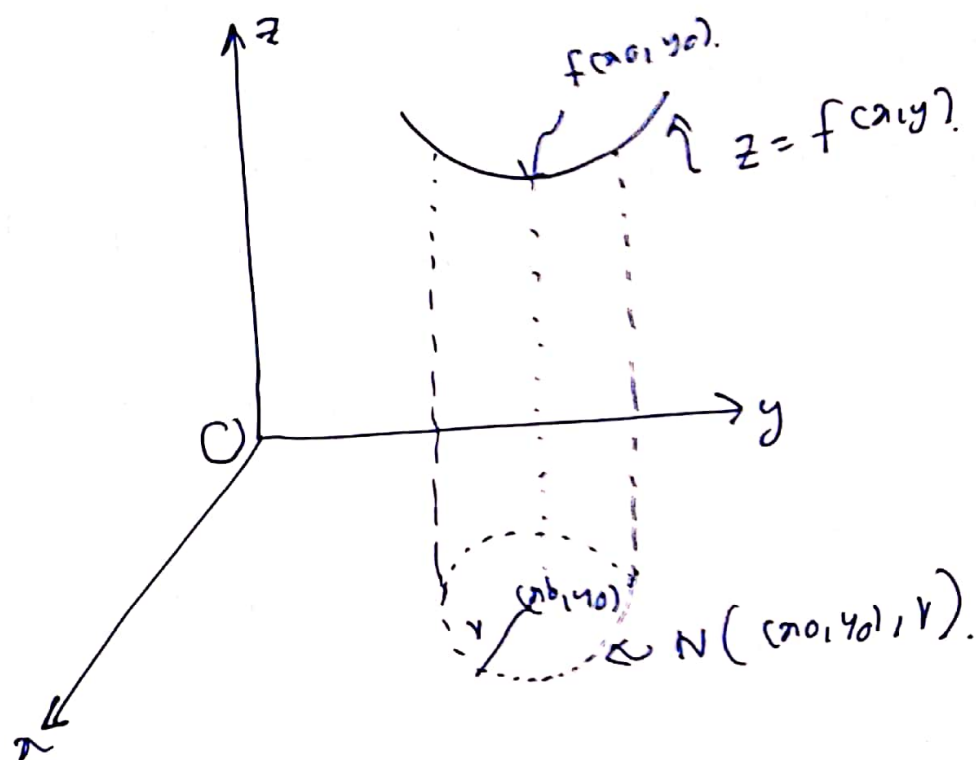


$$N((x_0, y_0), r) = \left\{ (x, y) \in \mathbb{R}^2 / 0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < r \right\}$$

i.e., f has a local maximum at (x_0, y_0) if $\exists r > 0$ such that

$$f(x, y) \leq f(x_0, y_0) \text{ for all } (x, y) \in N((x_0, y_0), r).$$

A local minimum is defined in a similar way.



i.e., f has a local minimum at (x_0, y_0) if $\exists \delta > 0$ such that

$$f(x_0, y_0) \leq f(x, y) \text{ for all } (x, y) \in N((x_0, y_0), \delta)$$

Definition

A point (x_0, y_0) where $\frac{\partial f(x_0, y_0)}{\partial x} = 0 = \frac{\partial f(x_0, y_0)}{\partial y}$ or at least one of $\frac{\partial f(x_0, y_0)}{\partial x}$ and $\frac{\partial f(x_0, y_0)}{\partial y}$ is undefined is called a critical point of f .

Definition

A function $z = f(x, y)$ has a saddle point at (x_0, y_0) if (x_0, y_0) is a critical point of f and for each $\epsilon > 0$ there exist points (x_1, y_1) and (x_2, y_2) in $N((x_0, y_0), \epsilon)$ with

$$f(x_1, y_1) < f(x_0, y_0) \text{ and } f(x_2, y_2) > f(x_0, y_0).$$

Example

Let $f(x, y) = y^2 - x^2$.

Then we have

$$\frac{\partial f}{\partial x} = -2x, \quad \frac{\partial f}{\partial y} = 2y.$$

The critical points of f are given by

$$\frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0.$$

i.e., $-2x = 0$ and $2y = 0$.

$$\Rightarrow x = 0 \text{ and } y = 0.$$

So, $(0, 0)$ is a critical point of f .

Is $(0, 0)$ a saddle point of f ?

~~Take $\epsilon > 0$.~~

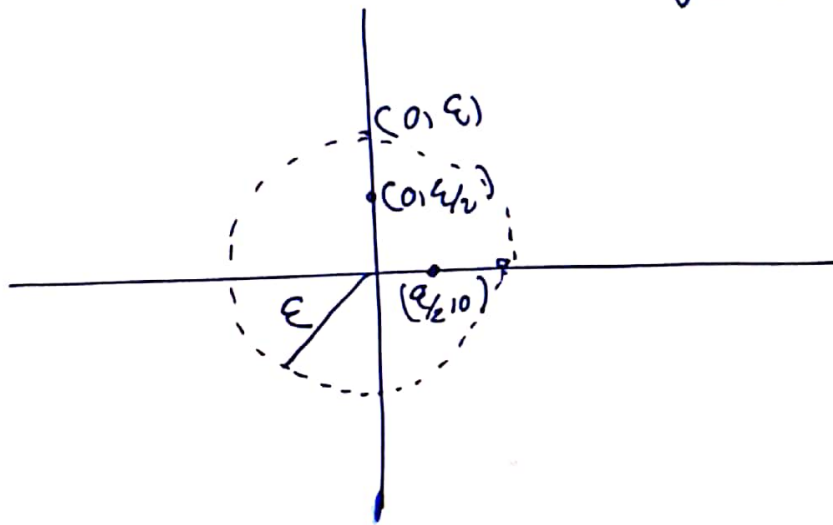
Take any $\epsilon > 0$.

If there exist $(x_1, y_1), (x_2, y_2) \in N((0,0), \epsilon)$
Such that

$$f(x_1, y_1) < f(0,0) \text{ and } f(x_2, y_2) > f(0,0),$$

then $(0,0)$ is a saddle point of f .

$$N((0,0), \epsilon) = \left\{ (x,y) \in \mathbb{R}^2 / 0 < \underbrace{\sqrt{(x-0)^2 + (y-0)^2}}_{\sqrt{x^2 + y^2}} < \epsilon \right\}.$$



$$f(0,0) = 0.$$

Note that $(\epsilon/2, 0), (0, \epsilon/2) \in N((0,0), \epsilon)$.

Take $(x_1, y_1) = (\epsilon/2, 0)$ and $(x_2, y_2) = (0, \epsilon/2)$.

$$f(x_1, y_1) = 0^2 - (\epsilon/2)^2 = -\frac{\epsilon^2}{4} < 0 = f(0,0)$$

$$f(x_2, y_2) = (\epsilon/2)^2 - 0^2 = \frac{\epsilon^2}{4} > 0 = f(0,0).$$

Therefore, f has a saddle point at $(0,0)$.

Suppose that the 2nd order partial derivatives of f are continuous in a disk centered at (x_0, y_0) and that

$$\frac{\partial f(x_0, y_0)}{\partial x} = 0 = \frac{\partial f(x_0, y_0)}{\partial y}.$$

$$\text{Let } A = \frac{\partial^2 f(x_0, y_0)}{\partial x^2},$$

$$B = \frac{\partial^2 f(x_0, y_0)}{\partial x \partial y},$$

$$C = \frac{\partial^2 f(x_0, y_0)}{\partial y^2}, \text{ and}$$

$$\Delta = AC - B^2.$$

$\begin{bmatrix} A & B \\ B & C \end{bmatrix}$ is called Hessian of f at (x_0, y_0) .

a) If $\Delta > 0$ and $A > 0$, then f has a local minimum at (x_0, y_0) .

b) If $\Delta > 0$ and $A < 0$, then f has a local maximum at (x_0, y_0) .

6

c) If $\Delta < 0$, then f has a saddle point at (x_0, y_0) .

Note:

If $\Delta = 0$, no conclusion can be drawn.
i.e., the test provides no information.

Example.

Let $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$.
Find and classify the critical points of f .

Solution

The critical points of f are given by

$$\frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0.$$

$$\Rightarrow 3x^2 + 3y^2 - 6x = 0 \text{ and } 6xy - 6y = 0.$$

$$\Rightarrow x^2 + y^2 - 2x = 0 \text{ and } xy - y = 0$$

$$\Rightarrow x^2 + y^2 - 2x = 0 \text{ and } y(x-1) = 0.$$

Case 1

$$x^2 + y^2 - 2x = 0 \text{ and } y = 0$$

$$\Rightarrow x^2 - 2x = 0 \Rightarrow x(x-2) = 0 \Rightarrow x = 0 \text{ or } x = 2.$$

Then the critical points of f are $(0,0)$, $(2,0)$.

Case 2

$$x^2 + y^2 - 2x = 0 \text{ and } x = 1$$

$$\Rightarrow 1 + y^2 - 2 = 0$$

$$\Rightarrow y^2 - 1 = 0$$

$$\Rightarrow (y-1)(y+1) = 0$$

$$\Rightarrow y = 1 \text{ or } y = -1.$$

Then the critical points of f are $(1,1)$, $(1,-1)$.

All the critical points of f are $(0,0)$, $(2,0)$, $(1,1)$, and $(1,-1)$.

At a point (x,y) ,

$$\begin{aligned} A = \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (3x^2 + 3y^2 - 6x) \\ &= 6x - 6 = 6(x-1). \end{aligned}$$

$$B = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (6xy - by) = 6y$$

$$C = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (6xy - by) = 6(x-1)$$

Critical points	A	B	C	$\Delta = AC - B^2$	Nature of the critical point
(0,0)	-6	0	-6	$3b > 0$	Local maximum
(2,0)	6	0	6	$3b > 0$	Local minimum
(1,1)	0	6	0	$-3b < 0$	Saddle point
(1,-1)	0	-6	0	$-3b < 0$	Saddle point

H.W.

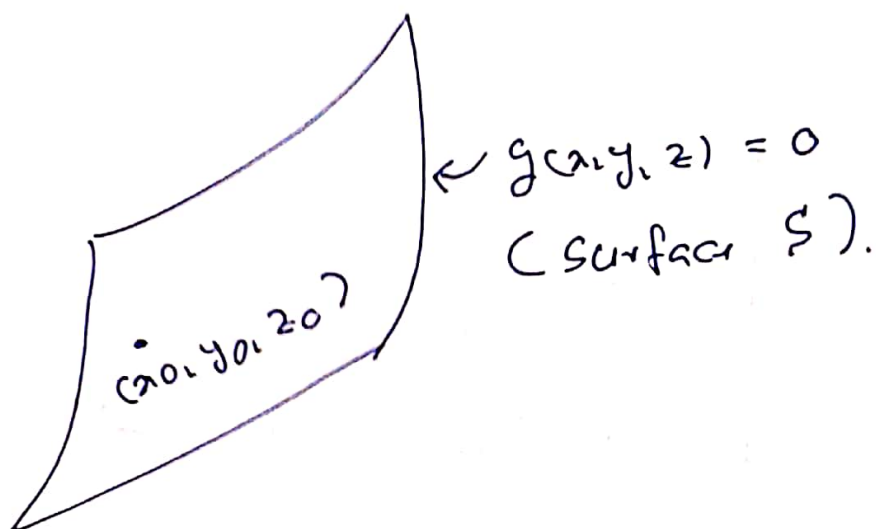
Let $f(x,y) = 8x^4 + 12x^2y - 9x^3$

Find and classify the critical points of f .

— Lagrange's Multipliers —

This is a method finding maxima/minima of a function of several variables subject to a constrain (a side condition).

Suppose that we have to find the extrema (Minima or maxima) of $f = f(x, y, z)$ subject to the condition (constrain) $g(x, y, z) = 0$.



Suppose that f has an extrema at (x_0, y_0, z_0) on S .

$$\text{Let } F = f + \lambda g.$$

Then we may show that

$$\nabla F(x_0, y_0, z_0) = \underline{0}$$

$$\Rightarrow \nabla(f+g\lambda)(x_0, y_0, z_0) = \underline{0}$$

$$\Rightarrow \frac{\partial(f+g\lambda)}{\partial x} \underline{i} + \frac{\partial(f+g\lambda)}{\partial y} \underline{j} + \frac{\partial(f+g\lambda)}{\partial z} \underline{k} = \underline{0}.$$

$$\Rightarrow (f_x + \lambda g_x) \underline{i} + (f_y + \lambda g_y) \underline{j} + (f_z + \lambda g_z) \underline{k} = \underline{0}.$$

Since $\underline{i}, \underline{j}$, and $\underline{k}$ are linearly independent,
we have

$$f_x + \lambda g_x = 0$$

$$f_y + \lambda g_y = 0$$

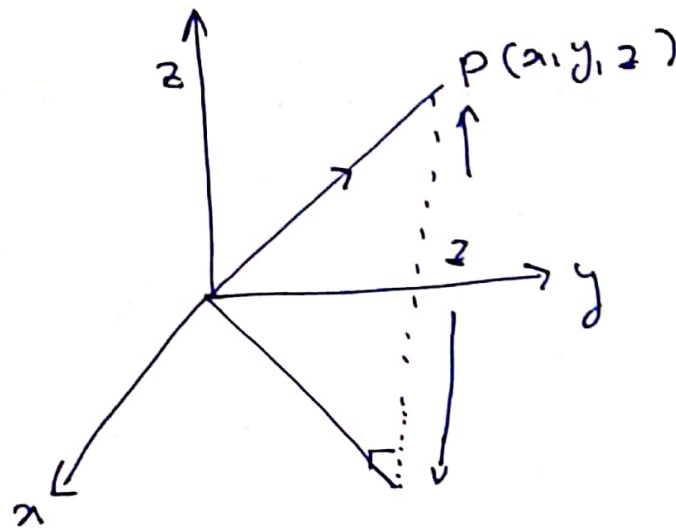
$$f_z + \lambda g_z = 0 \text{ at}$$

at (x_0, y_0, z_0) and $g(x_0, y_0, z_0) = 0$.

Example

Find the minimum value of $f(x, y, z) = x^2 + y^2 + z^2$
Subject to $2x + 3y - z = 1$.

Solution



$$2x + 3y - z = 1 :$$

On zoy -plane ($x=0$);

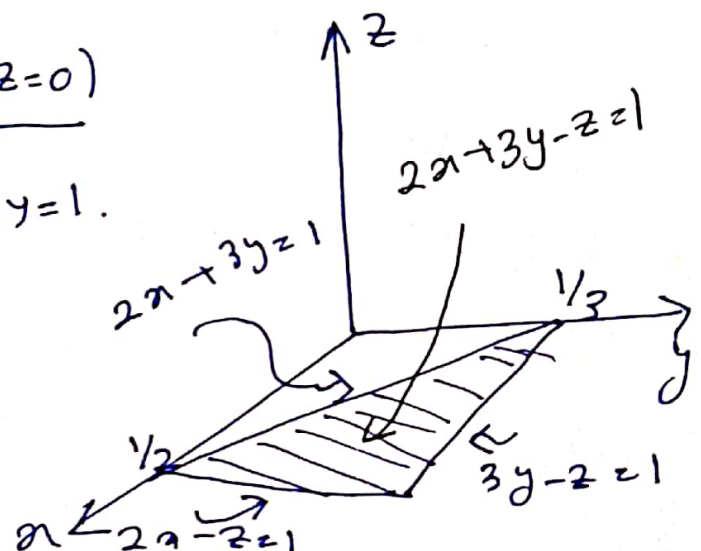
$$x=0 \Rightarrow 3y - z = 1$$

On xoz -plane ($y=0$);

$$y=0 \Rightarrow 2x - z = 1$$

On xoy -plane ($z=0$)

$$z=0 \Rightarrow 2x + 3y = 1.$$



Let $g(x, y, z) = 2x + 3y - z - 1$

$f(x, y, z) = x^2 + y^2 + z^2 = (\text{distance})^2$ from the origin is to be minimized assuming that (x, y, z) lies on $g(x, y, z) = 0$.

Maximum does not exist but the minimum exists.

Let $F = f + \lambda g$ for some scalar λ ($\neq 0$).

$$\Rightarrow F = x^2 + y^2 + z^2 + \lambda(2x + 3y - z - 1)$$

$$\Rightarrow F_x = 2x + 2\lambda$$

$$F_y = 2y + 3\lambda$$

$$F_z = 2z - \lambda$$

Now by Lagrange's multipliers, at the minimum of f (subject to $g(x, y, z) = 0$), we have:

$$F_x = 2x + 2\lambda = 0 \Rightarrow x = -\lambda \quad \text{--- (1)}$$

$$F_y = 2y + 3\lambda = 0 \Rightarrow y = -3\lambda/2 \quad \text{--- (2)}$$

$$F_2 = 2z - 1 = 0 \Rightarrow z = 1/2. \text{---} (3)$$

We also have

$$g(x, y, z) = 2x + 3y - z - 1 = 0 \text{---} (4)$$

Substituting x , y , and z from (1), (2), and (3), respectively into (4), we have:

$$2(-1) + 3(-3/2) - 1/2 - 1 = 0$$

$$-2 - 9/2 - 1/2 - 1 = 0$$

$$\therefore 1 = -1/7$$

So, $x = 1/7$, $y = -3/2(-1/7) = 3/14$ and $z = -1/14$.

Therefore, the minimum of f must occur at $(1/7, 3/14, -1/14)$.

The minimum value of f

$$= f(1/7, 3/14, -1/14)$$

$$= (1/7)^2 + (3/14)^2 + (-1/14)^2 = 0.129$$

Example

Suppose that the temperature at a point (x, y, z) on the unit sphere $S: x^2 + y^2 + z^2 = 1$ is given by $T(x, y, z) = 30 + 5(x + z)$.

Find the ~~extreme~~ extreme values of T .

Solution

Let $F = 30 + 5(x + z) + \lambda(x^2 + y^2 + z^2 - 1)$,

where $\lambda \neq 0$.

We then have

$$F_x = 5 + 2\lambda x,$$

$$F_y = 2\lambda y$$

$$F_z = 5 + 2\lambda z.$$

Now by Lagrange's multipliers, at the extreme values of T , we have:

$$5 + 2\lambda x = 0 \quad \text{--- (1)}$$

$$2\lambda y = 0 \quad \text{--- (2)}$$

$$5 + 2\lambda z = 0 \quad \text{--- (3)}$$

We also have:

$$x^2 + y^2 + z^2 = 1 \quad \text{--- (4)}$$

$$(1) \Rightarrow x = -\frac{5}{2\gamma}$$

$$(2) \Rightarrow y = 0$$

$$(3) \Rightarrow z = -\frac{5}{2\gamma}, \text{ since } \gamma \neq 0.$$

Now (1) and (3) give us

Then (4) gives

$$\left(-\frac{5}{2\gamma}\right)^2 + 0^2 + \left(-\frac{5}{2\gamma}\right)^2 = 1 \Rightarrow \gamma = \pm \frac{\sqrt{2}}{5} \cdot \frac{5}{\sqrt{2}}.$$

Then $x = \pm \frac{1}{\sqrt{2}}$, $y = 0$ and $z = \pm \frac{1}{\sqrt{2}}$.

i.e., the extreme points are $\left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ and $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$.

$$\begin{aligned} \text{When At } \left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), T &= 30 + 5\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) \\ &= 30 - 5\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{and at } \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), T &= 30 + 5\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) \\ &= 30 + 5\sqrt{2}. \end{aligned}$$

HW

Find the maximum and minimum values of $x^2 + y^2 + z^2$ Subject to the conditions:

$$\frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} = 1 \text{ and } z = x + y.$$

Let $g(x, y, z) = \frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} - 1$ and

$$h(x, y, z) = x + y - z.$$

Let $F = f + \lambda g + \mu h$

$$= x^2 + y^2 + z^2 + \lambda \left(\frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} - 1 \right) + \mu (x + y - z),$$

when $\lambda, \mu \neq 0$.